Metadata for Bude (BUD), Cornwall, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field and the field created by GIC at the underline site (BUDU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station (BUDU)	50.763149 N, -4.425513E
Location (in lat/long): remote station	N/A (equipment failure at remote after installation)
Description of site (i.e. open moorland)	Open sloping field with cattle. Underline was fenced in to prevent damage.
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	26/02/2020
Removal date	07/10/2020
Number of days collected	223
Data sampling rate	1 Hz
Notes of known data gaps	No remote data available
Qualitative estimate of site quality	Good
Orientation angle of HV wires relative to true north	+44.86° (from True North)
Alignment of magnetometers	Magnetic north as of 26/02/2020
Voltage level of the lines	400 kV (Indian Queens – Alverdiscott)
Height above the ground of each wire (m)	Lower: 16.98 m, Middle: 25.40 m, Upper: 35.31 m
Underline magnetometer	Placed under northern circuit, 18m to second circuit
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421